

US Patent & Trademark Office

Patent Public Search | Text View

United States Patent	12404839
Kind Code	B2
Date of Patent	September 02, 2025
Inventor(s)	Nauheimer; Michael et al.

Wind turbine blade and wind turbine with a down conductor spar cap

Abstract

A wind turbine blade for a wind turbine is provided, the wind turbine blade having a tip portion and a root portion, and the wind turbine blade including a shell and a spar, the spar including two spar caps connected to one another by at least one spar web of the spar and extending in a longitudinal direction of the wind turbine blade, whereby at least one of the two spar caps is configured as a main down conductor of a lightning protection system of the wind turbine blade, the at least one main down conductor including an electrically conductive fiber-reinforced plastic being electroconductively connected to at least one electrical interface of the lightning protection system. Also provided is a wind turbine including such a wind turbine blade.

Inventors:	Nauheimer; Michael (Aalborg, DK), Ponnada; Sri Markandeya Rajesh (Boucherville, CA)
Applicant:	Siemens Gamesa Renewable Energy A/S (Brande, DK)
Family ID:	75870602
Assignee:	SIEMENS GAMESA RENEWABLE ENERGY A/S (Brande, DK)
Appl. No.:	17/922391
Filed (or PCT Filed):	May 03, 2021
PCT No.:	PCT/EP2021/061585
PCT Pub. No.:	WO2021/228606
PCT Pub. Date:	November 18, 2021

Prior Publication Data

Document Identifier	Publication Date
US 20230349364 A1	Nov. 02, 2023

Foreign Application Priority Data

IN 202031020283 May. 14, 2020

Publication Classification

Int. Cl.: F03D80/30 (20160101); F03D1/06 (20060101)

U.S. Cl.:

CPC F03D80/30 (20160501); F03D1/0675 (20130101); F03D1/0679 (20230801); F03D1/0681 (20230801); F03D80/301 (20230801); F05B2280/2006 (20130101); F05B2280/6003 (20130101)

Field of Classification Search

USPC: None

References Cited

U.S. PATENT DOCUMENTS

Patent No.	Issued Date	Patentee Name	U.S. Cl.	CPC
6457943	12/2001	Olsen	416/241A	F03D 80/30
7494324	12/2008	Hibbard	416/241A	H02G 13/00
7729100	12/2009	Llorente Gonzalez	361/220	B82Y 30/00
8118559	12/2011	Llorente Gonzalez	416/226	F03D 1/065
8657582	12/2013	Hibbard	416/229R	F03D 80/30
9689377	12/2016	Klein	N/A	F03D 80/40
9719495	12/2016	Chacon	N/A	F03D 13/20
10125744	12/2017	March Nomen	N/A	F03D 80/30
10294925	12/2018	Klein	N/A	F03D 1/0675
10330075	12/2018	March Nomen	N/A	F03D 80/30
10648456	12/2019	Shain	N/A	F03D 80/30
11994112	12/2023	Hayden	N/A	H02G 13/40
2007/0074892	12/2006	Hibbard	N/A	N/A
2008/0073098	12/2007	Llorente Gonzalez	174/2	B82Y 30/00
2012/0134826	12/2011	Arocena De La Rua et al.	N/A	N/A
2014/0241896	12/2013	Zhang et al.	N/A	N/A
2015/0292479	12/2014	Ohlerich et al.	N/A	N/A
2016/0138569	12/2015	Caruso et al.	N/A	N/A

2016/0327028	12/2015	March Nomen et al.	N/A	N/A
2016/0333861	12/2015	Chacon	N/A	F03D 80/30
2018/0156202	12/2017	Lipka et al.	N/A	N/A
2018/0245566	12/2017	Sawada	N/A	F03D 1/0675

FOREIGN PATENT DOCUMENTS

Patent No.	Application Date	Country	CPC
103797243	12/2013	CN	N/A
205805824	12/2015	CN	N/A
19748716	12/1997	DE	B64C 27/473
2930355	12/2014	EP	N/A
3 255 275	12/2016	EP	N/A
3483424	12/2018	EP	N/A
3501808	12/2018	EP	N/A
3501809	12/2018	EP	N/A
3501810	12/2018	EP	N/A
3511560	12/2018	EP	N/A
00/14405	12/1999	WO	N/A
WO-2017089591	12/2016	WO	F03D 1/0675
2018/101632	12/2017	WO	N/A

OTHER PUBLICATIONS

English translation of WO2017/089591A1 (Year: 2017). cited by examiner

English translation of DE19748716C1 (Year: 1998). cited by examiner

International Search Report and Written Opinion for PCT/EP2021/061585 issued on Aug. 4, 2021.
cited by applicant

Primary Examiner: Lee, Jr.; Woody A

Assistant Examiner: Fisher; Wesley Le

Attorney, Agent or Firm: Schmeiser, Olsen & Watts LLP

Background/Summary

CROSS-REFERENCE TO RELATED APPLICATIONS

(1) This application claims priority to PCT Application No. PCT/EP2021/061585, having a filing date of May 3, 2021, which claims priority to Indian Application No. 202031020283, having a filing date of May 14, 2020, the entire contents both of which are hereby incorporated by reference.

FIELD OF TECHNOLOGY

(2) The following relates to a wind turbine blade having an improved lightning protection system and a wind turbine having an improved lightning protection system.

BACKGROUND

(3) Lightning is likely to strike wind turbines due to their exposed location and height. As lightning strikes may cause considerable damage to a wind turbine, wind turbines are typically protected by a lightning protection system. The objective of the lightning protection system is to prevent lightning

from damaging the wind turbine blades, bearings and electrical systems of the wind turbine when a wind turbine blade intercepts a lightning strike.

(4) In particular, the wind turbine blades of the wind turbines are susceptible to both interception of lightning strikes when they are at a pointer position of or near 12 o'clock, i.e., showing up to the sky, and taking severe damage from lightning strikes. For this purpose, the wind turbine blades are provided with metallic down conductors as their main down conductors and part of their lightning protection system. The metallic down conductor is attached to a spar web of a spar of the wind turbine blade extending in a longitudinal direction of the wind turbine blade. Thereby, the metallic down conductor in the wind turbine blade extends from its tip to its root at which the wind turbine blade is attached to a hub of the wind turbine. The lightning protection system of the wind turbine further comprises down conductors running from the hub via the nacelle down to the tower, acting as a further down conductor to earth or having down conductors inside or outside of the tower and being grounded.

(5) When the spar caps of the spar of the wind turbine blade having the metallic down conductor comprise or are made from electrically conductive fiber-reinforced polymer equipotential bonds between the spar caps and the metallic down conductor need to be provided. For example, carbon-fiber-reinforced plastic is increasingly used in the spar caps due to its high stiffness-to-mass ratio, thereby enhancing the performance of the wind turbine blades. The equipotential bonds in spar caps employing an electroconductive fiber-reinforced plastic are necessary to ensure that a voltage difference due to a current flow from a lightning strike through the spar caps or the metallic down-conductor with respect to the metallic down conductor is under a certain threshold for preventing a flashover, which may lead to structural damage of the wind turbine blade.

(6) Further, such equipotential bonds typically go through high-strain regions of the wind turbine blade and thus the spar caps cannot be electroconductively connected to the metallic down conductor in a straight manner. Thereby, the equipotential bonds reduce the design flexibility of the wind turbine blades.

(7) Moreover, the metal-conductive fiber interface between the electrically conductive fiber-reinforced plastic and the metallic down conductor can form a galvanic couple, thereby possibly leading to corrosion. Also, impedance difference between the plastic, such as carbon, and the metal of the metallic down conductor, such as copper, leads to a high risk of delamination of the spar caps, due to the high current flow resulting from the lightning strike.

(8) Also, the equipotential bonds are typically manufactured manually, and their manufacturing quality may thus vary. However, manufacturing defects are difficult to detect, and accessing and repairing them in the installed wind turbine is very complex.

(9) Ultimately, all equipotential bonds and the metallic down conductor add weight and material costs as well as manufacturing costs and time to the manufacturing process of the wind turbine blade. Also, the attachment of the metallic down conductor in the wind turbine blade, typically at the spar web of the spar, poses further manufacturing steps, additional material and risk of defects.

SUMMARY

(10) An aspect relates to a wind turbine blade having an improved lightning protection system not having the disadvantages from the state of the art, whereby in particular the wind turbine blade may be manufactured with less manufacturing steps, material and costs and still provide the necessary lightning protection capabilities to avoid damage due to lightning strikes.

(11) Thereby, the features and details described in connection with the wind turbine blade of embodiments of the invention apply in connection with the wind turbine, so that regarding the disclosure of the individual aspects of embodiments of the invention it is or can be referred to one another.

(12) According to a first aspect of embodiments of the invention, the object is solved by a wind turbine blade for a wind turbine, the wind turbine blade having a tip portion and a root portion, and the wind turbine blade comprising a shell and a spar, the spar comprising two spar caps connected

to one another by at least one spar web of the spar and extending in a longitudinal direction of the wind turbine blade, whereby at least one of the two spar caps is configured as a main down conductor of a lightning protection system of the wind turbine blade, the at least one main down conductor comprising an electrically conductive fiber-reinforced plastic being electroconductively connected to at least one electrical interface of the lightning protection system.

(13) The wind turbine blade according to the first aspect of embodiments of the invention lets go of the solution of a metallic down conductor as a main down conductor of its lightning protection system as known in the state of the art. Instead, at least one of the two spar caps of the spar comprising electrically conductive fiber-reinforced plastic is configured as the main down conductor. The electroconductive properties of the material of the spar cap and its large surface area are thereby favorably utilized for conducting current flow from an intercepted lightning strike. Thereby, the usage of electrically conductive fiber-reinforced plastic in the spar caps accepted as a disadvantage over usage of non-electroconductive fiber-reinforced plastic in the spar caps in the state of the art, i.e., resulting in the necessity of equipotential bonds with the metallic down conductor possibly resulting in corrosion, defects and high manufacturing costs and time, is turned into an advantage. There are no equipotential bonds of a metal-conductive fiber type with a metallic down conductor anymore and hereby the risk of corrosion is eliminated, and the risk of defects and the costs of manufacturing are significantly decreased.

(14) It may be preferred that the at least one main down conductor is configured to mainly conduct electricity from a lightning strike from the tip portion to the root portion of the wind turbine blade. According to this feature, when both spar caps are configured as main down conductors, both main down conductors are configured to mainly conduct electricity from a lightning strike from the tip portion to the root portion of the wind turbine blade. Mainly means that more than half of the current of an intercepted lightning strike is conducted via the at least one main down conductor being the at least one spar cap. This may typically not be achieved when a metallic down conductor is provided over the length of the wind turbine blade, because the electroconductivity of the metallic down conductor will in most if not all cases be greater than the electroconductivity of electrically conductive fiber-reinforced plastic of the spar caps. In particular, the at least one main down conductor may be configured to entirely or substantially entirely conduct electricity from a lightning strike from the tip portion to the root portion of the wind turbine blade.

(15) It may also be preferred, that the electrically conductive fiber-reinforced plastic of the at least one main down conductor is electroconductively connected to a first electrical interface at the tip portion and electroconductively connected to a second electrical interface at the root portion. The first electrical interface and the second electrical interface are of the at least one electrical interface of the lightning protection system. Thereby, the at least one main down conductor extends from the tip portion to the root portion and is fully integrated into the lightning protection system.

(16) Therein, it may be preferred that the first electrical interface is an electrical air-termination arrangement and/or the second electrical interface is an electrical root terminal. The air-termination arrangement may in particular comprise an air-termination base and at least one or multiple air-terminations mounted on the air-termination base. The root terminal may be an electrical terminal at the root of the wind turbine blade for electroconductive connection to the hub of the wind turbine. The electrical air-termination arrangement may be arranged at the tip or in a tip portion of the wind turbine blade. Also, there may be at least one further or multiple first electrical interfaces electroconductively connected to the electrically conductive fiber-reinforced plastic of the main down conductor. The at least one further or the multiple first electrical interfaces may be further electrical air-termination arrangements. The electrical air-termination arrangement may intercept a lightning strike at the tip portion of the wind turbine blade. The electrical root terminal may be electroconductively connected to the lightning protection system of the wind turbine, in particular at its hub.

(17) It may also be preferred that each of the two spar caps is configured as a main down conductor

comprising an electrically conductive fiber-reinforced plastic being electroconductively connected to the at least one electrical interface of the lightning protection system. Thereby, the electroconductive properties of both spar caps are favorably used for the purpose of conducting current from a lightning strike.

(18) Moreover, it may be preferred that the at least one main down conductor predominantly comprises or is made from electrically conductive fiber-reinforced plastic. Thereby, its electroconductive properties are enhanced.

(19) Also, it may be preferred that the electrically conductive fiber-reinforced plastic of the at least one main down conductor is carbon-fiber-reinforced plastic. Carbon-fiber-reinforced plastic provides for particularly good electrical conductivity and has a favorable stiffness-to-mass ratio for employment in the spar caps. The electrically conductive fiber-reinforced plastic may be further reinforced with at least one other material that may or may not be in the form of a fiber. This material may have high electroconductive properties and/or high mechanical strength and/or stiffness to improve electrical and/or mechanical properties of the spar caps and thereby the wind turbine blade.

(20) Further, it may be preferred, that the two spar caps are connected by at least one equipotential bond connection arranged in between the tip portion and the root portion. The equipotential bond connection reduces the risk of flashovers from one spar cap to the other one and thereby the risk of structural damage to the spar caps due to lightning strikes.

(21) Therein, it may be preferred that multiple equipotential bond connections connect the two spar caps in between the tip portion and the root portion, whereby the multiple equipotential bond connections are arranged at distance from one another along the longitudinal direction of the wind turbine blade. Thereby, the risk of flashovers is further reduced. This may be possible by having less equipotential bond connections than equipotential bonds in the state of the art, because the distance between the spar caps is greater compared to the distance between a spar cap and the metallic down conductor as known in the state of the art.

(22) Therein, it may further be preferred that the at least one equipotential bond connection predominantly comprises or is made from electrically conductive fiber-reinforced plastic. In particular, the electrically conductive fiber-reinforced plastic of the equipotential bond connection may be the same as the electrically conductive fiber-reinforced plastic of the at least one main down conductor. Thereby, an electrically conductive fiber reinforced plastic—electrically conductive fiber reinforced plastic interface is provided. Such an interface is easier to manufacture than the metal-conductive fiber interface for the equipotential bonds as known in the state of the art and does not have the disadvantages, such as risk of corrosion and/or impedance mismatch associated therewith.

(23) Also, it may be preferred that the at least one electrical interface predominantly comprises metal or is made from metal. For example, copper or a metal alloy having copper may be used as the metal. Thereby, at the at least one electrical interface, a particularly high electrical conductivity is provided to conduct the current from the lightning strike.

(24) Further, it may be preferred that the at least one electrical interface is electroconductively connected to the electrically conductive fiber-reinforced plastic of the at least one main down conductor by a metallic conductor connection predominantly comprising metal or being made from metal. For example, copper or a metal alloy having copper may be used as the metal. Thereby, at the connection between the at least one main down conductor and the at least one electrical interface, a particularly high electrical conductivity is provided to conduct the current from the lightning strike.

(25) In particular, it may be preferred that a longitudinal portion of the wind turbine blade in between the tip portion and the root portion and in between the two spar caps does not comprise a metallic down conductor predominantly comprising or being made from metal. Thereby, the disadvantages associated with integration of a metallic down conductor in the lightning protection

system of the wind turbine blade are avoided.

(26) Also, it may be preferred that the spar web or the wind turbine blade does not comprise a metallic down conductor predominantly comprising metal or being made from metal. In particular, the spar web or the wind turbine blade may not comprise a further down conductor besides the one or two of the two spar caps configured as main down conductors. In other words, the one or two of the two spar caps configured as main down conductors may be the only down conductors of the wind turbine blade. Thereby, the disadvantages associated with integration of a metallic down conductor in the lightning protection system of the wind turbine blade are avoided.

(27) According to a second aspect of embodiments of the invention, the aspect is solved by a wind turbine comprising at least one wind turbine blade according to the first aspect of embodiments of the invention, whereby the lightning protection system of the at least one wind turbine blade is electroconductively connected to a lightning protection system of the wind turbine.

(28) Further advantages, features and details of embodiments of the invention unfold from the following description, in which by reference to the drawings an embodiment of the state of the art and an embodiment of the present invention are described in detail. Thereby, the features from the claims as well as the features mentioned in the description can be essential for the invention as taken alone or in an arbitrary combination.

Description

BRIEF DESCRIPTION

(1) Some of the embodiments will be described in detail, with reference to the following figures, wherein like designations denote like members, wherein:

(2) FIG. 1 shows a side perspective view on an embodiment of a wind turbine;

(3) FIG. 2 shows a sectional cut through a wind turbine blade of the wind turbine of FIG. 1 and according to an embodiment of the state of the art;

(4) FIG. 3 shows a schematic view on the lightning protection system of the wind turbine blade of FIG. 2;

(5) FIG. 4 shows a flashover in the wind turbine blade of FIG. 3;

(6) FIG. 5 shows a schematic view on the lightning protection system of a wind turbine blade according to an embodiment of the invention; and

(7) FIG. 6 shows absence of flashover in the wind turbine blade of FIG. 5.

(8) Same objects in FIGS. 1 to 6 are denominated with the same reference number. If there is more than one object of the same kind in one of the figures, the objects are numbered in ascending order with the ascending number of the object being separated from its reference number by a dot. The specific dimensions of features and parts in the figures are exemplary and may be enlarged for ease of reference only.

DETAILED DESCRIPTION

(9) The wind turbine **1** as shown in FIG. 1 comprises a tower **2**, a nacelle **3** and a hub **4**. The nacelle **3** is supported on the tower **2** and the hub **4** is attached to the nacelle **3**. In this exemplary case, three wind turbine blades **10.1**, **10.2**, **10.3** are attached to the hub **4**. However, the number of wind turbine blades **10** may alternatively be less or more than three. As exemplary shown for the wind turbine blade **10.1**, the wind turbine blade **10.1** extends in a longitudinal direction **L** from a root portion **15** to a tip portion **14** thereof. The features of the wind turbine **1** as mentioned above may be employed in a wind turbine **1** according to embodiments of the invention.

(10) The wind turbine blade **10.1** of the wind turbine **1** of FIG. 1 is cut along line II-II. The sectional cut along line II-II for a wind turbine blade **10** according to an embodiment of the state of the art is illustrated in FIG. 2.

(11) The wind turbine blade **10** comprises a shell **11** and a spar. The spar comprises two spar caps

12.1, 12.2 located opposite of one another and being connected to each other by a spar web **13**. With the above-mentioned features, the wind turbine blade **10** shown in FIG. 2 may be the same as one according to an embodiment of the invention. However, different from such an embodiment of the invention, the wind turbine blade **10** as shown in FIG. 2 comprise a metallic down conductor **20** as a main down conductor. The metallic down conductor **20** is attached to the spar web **13** and runs in the longitudinal direction L of the wind turbine blade **10** from its tip portion **14** to its root portion **15**.

(12) FIG. 3 shows a schematic view on the lightning protection system of the wind turbine blade **10** of FIG. 2. The spar caps **12.1, 12.2** are made from an electrically conductive fiber-reinforced plastic. The metallic down conductor **20** extends in the longitudinal direction L of the blade and is connected to a first electrical interface **22** at the tip portion **14** of the wind turbine blade **10** and to a second electrical interface **23** at the root portion **15** of the wind turbine blade **10**. Equipotential bond connections **21**, one of which is exemplary denominated, are provided with distance from one another along the wind turbine blade **10** to prevent flashovers F when current from a lightning strike flows through the spar caps **12.1, 12.2** and the metallic down conductor **20**. The equipotential bond connections **21** are formed by multiple metallic connectors extending transverse, in particular substantially perpendicular, to the metallic down conductor **20**.

(13) A cut-out III from FIG. 3 is shown in detail in FIG. 4, where a flashover F occurs in between the spar cap **12.1** and the metallic down conductor **20**. This may occur when the number of equipotential bond connections **21** is low and the current from the lightning strike is very strong. This may lead to structural damage of the wind turbine blade **10** and must be prevented by providing an adequate, high number of equipotential bond connections **21**. However, these equipotential bond connections **21** are of a metal-conductive fiber type and thereby lead to several disadvantages such as reduced design flexibility of the wind turbine blades, possibly corrosion at the metal-conductive fiber interface, delamination in or around the spar caps and manufacturing defects as explained in detail in the introductory part of this specification.

(14) As shown in the lightning protection system of the wind turbine blade **10** according to an embodiment of the invention in FIG. 5, the metallic down conductor **20** is omitted. Instead, the spar caps **12.1, 12.2** are utilized as main down conductors **24.1, 24.2**. The main down conductors **24.1, 24.2** are the only down conductors of the wind turbine blade **10**. At the tip portion **14** of the wind turbine blade **10**, the main down conductors **24.1, 24.2** are connected to at least one first electrical interface, in this particular case two first electrical interfaces **22.1, 22.2**, via a first metallic conductor connection **31.1**. In this case, the first electrical interfaces **22.1, 22.2** are provided as electrical air-terminations **22.1, 22.2** of an electrical air-termination arrangement **22** in the tip portion **14** of the wind turbine blade **10**. At the root portion **15** of the wind turbine blade **10**, the main down conductors **24.1, 24.2** are connected to a second electrical interfaces **23** via a second metallic conductor connection **31.2**. In this case, the second electrical interface **23** is provided as an electrical root terminal **23**. Thus, the lightning protection system of the wind turbine blade **10** comprises a first segment as a first metallic conductor connection **31.1**, a second segment as the spar caps **12.1, 12.2** configured as main down conductors **24.1, 24.2** and a third segment configured as a second metallic conductor connection **31.2**. The first segment is connected to the second segment by a connection of the first metallic conductor connection **31.1** with an electrically conductive fiber connection **32.1** between the first spar cap **12.1** and the second spar cap **12.2**. The second segment is connected to the third segment by a connection of the second metallic conductor connection **31.2** with an electrically conductive fiber connection **32.2** between the first spar cap **12.1** and the second spar cap **12.2**. Instead of the electrically conductive fiber connections **32.1, 32.2**, the first segment and the second segment may be directly connected to each of the first spar cap **12.1** and the second spar cap **12.2**. The electrically conductive fiber-connections **32.1, 32.2** form a plastic-plastic connection with the spar caps **12.1, 12.2**. The material used for the electrically conductive fiber-connections **32.1, 32.2** may be a fiber-reinforced plastic, in particular

the same fiber-reinforced plastic as used for the spar caps **12.1**, **12.2**, for example carbon-fiber-reinforced plastic. At the first segment, being at the tip portion **14** of the wind turbine blade **10**, and at the third segment, being at the root portion **15** of the wind turbine blade **10**, a particularly high electrical conductivity is provided for favorably carrying the current from the lightning strike to the spar caps **12.1**, **12.2** and away from the spar caps **12.1**, **12.2**.

(15) In order to prevent flashovers F from occurring when a lightning strike is intercepted by one of the electrical air-terminations **22.1**, **22.2**, equipotential bond connections **30** may be provided in between the spar caps **12.1**, **12.2**. In FIG. 5, one equipotential bond connection **30** is shown, however, the number of the equipotential bond connections **30** may be adjusted as needed. Different from the equipotential bond connections **21** of the wind turbine blade **10** of FIG. 3, the equipotential bond connections **30** employed in the wind turbine blade **10** of FIG. 5 do have a plastic-plastic or conductive fiber-conductive fiber interface and thereby do not show the disadvantages associated with the metal-conductive fiber interface present in the wind turbine blade **10** of FIG. 3. The material used for the equipotential bond connection **30** may be a fiber-reinforced plastic, in particular the same fiber-reinforced plastic as used for the spar caps **12.1**, **12.2**, for example carbon-fiber-reinforced plastic. Alternatively, the equipotential bond connections **30** may employ metal together with electroconductive fiber-reinforced plastic, which may be provided as flaps or extensions connecting the spar caps **12.1**, **12.2** with the metal, for example.

(16) Moreover, the number of equipotential bond connections **30** employed in the wind turbine blade **10** of FIG. 5 may be less than in the wind turbine blade **10** of FIGS. 3 and 4. This is explained with reference to the cut-out IV of FIG. 5, which is shown in detail in FIG. 6. FIG. 6 shows a flashover F, which may occur in the wind turbine blade **10** of FIG. 5, from a lightning strike of same strength as in FIG. 4. However, different from FIG. 4, in FIG. 5 there is no metallic down conductor **20** but the spar cap **12.2** as the further main down conductor **24.2**. The two main down conductors **24.1**, **24.2** are separated from one another by a further distance D compared to the wind turbine blade **10** of FIGS. 3 and 4, where the metallic down conductor **20** is closer to the spar cap **12.1**. Thereby, the flashover F would have to be larger, i.e., bridge a greater distance between electroconductive objects, in order to discharge itself and possibly damage the wind turbine blade **10**. Due to the increased distance D, the likelihood of flashovers F and structural damage is decreased and hence the number of equipotential bond connections **30** may be less, effectively resulting in less material, weight, manufacturing costs and time.

(17) When the wind turbine blade **10** has a tapering geometry in the tip portion **14**, it may be provided that the spar caps **12.1** and **12.2** extend to join each other, thereby avoiding the need for equipotential connection **32.1**. Thus, such a tapered wind turbine blade **10** with joining spar caps **12.1**, **12.2** at its tip portion **14** may omit the equipotential connection **32.2** at the tip portion **14**.

(18) Also, the first metallic conductor connection **31.1** may be omitted when the first segment is omitted. Alternatively, the first metallic conductor connection **31.1** may be transverse to the longitudinal direction L, i.e., in the chordwise direction of the wind turbine blade **10**.

(19) Further, it is also possible that in the root portion **15** of the blade **10**, the third segment may comprise more than one metallic down-conductor **31.2** connected to the root terminal **23**, in particular two or more metallic down-conductors **31.2**. Similarly, the first segment may comprise more than one metallic down-conductor connection **31.1**.

(20) Although the present invention has been disclosed in the form of embodiments and variations thereon, it will be understood that numerous additional modifications and variations could be made thereto without departing from the scope of the invention.

(21) For the sake of clarity, it is to be understood that the use of “a” or “an” throughout this application does not exclude a plurality, and “comprising” does not exclude other steps or elements.

Claims

1. A wind turbine blade for a wind turbine, the wind turbine blade comprising: a tip portion; a root portion; a shell; a spar comprising two spar caps connected to one another by at least one spar web of the spar and extending in a longitudinal direction of the wind turbine blade, wherein, at least one of the two spar caps is configured as a main down conductor of a lightning protection system of the wind turbine blade, the at least one main down conductor comprising an electrically conductive fiber-reinforced plastic being electroconductively connected to at least one electrical interface of the lightning protection system, wherein the at least one electrical interface comprises a first electrical interface at the tip portion and a second electrical interface at the root portion; wherein the two spar caps are connected by a first equipotential bond connection and a second equipotential bond connection; wherein a first metallic conductor connection connects the first electrical interface to the first equipotential bond connection, and a second metallic conductor connection connects the second equipotential bond connection to the second electrical interface at the root portion; wherein the first equipotential bond connection and the second equipotential bond connection connect the two spar caps without contacting a metal down conductor arranged between the two spar caps along a thickness direction.
2. The wind turbine blade according to claim 1, wherein, the at least one main down conductor is configured to mainly conduct electricity from a lightning strike from the tip portion to the root portion of the wind turbine blade.
3. The wind turbine blade according to claim 1, wherein, the first electrical interface is an electrical air-termination arrangement and/or the second electrical interface is an electrical root terminal.
4. The wind turbine blade according to claim 1, wherein, each of the two spar caps is configured as a main down conductor comprising an electrically conductive fiber-reinforced plastic being electroconductively connected to the at least one electrical interface of the lightning protection system.
5. The wind turbine blade according to claim 1, wherein, the at least one main down conductor comprises or is made from electrically conductive fiber-reinforced plastic.
6. The wind turbine blade according to claim 1, wherein, the electrically conductive fiber-reinforced plastic of the at least one main down conductor is carbon-fiber-reinforced plastic.
7. The wind turbine blade according to claim 1, wherein the first equipotential bond connection and the second equipotential bond connection are arranged in between the two spar caps.
8. The wind turbine blade according to claim 7, wherein, multiple equipotential bond connections connect the two spar caps in between the tip portion and the root portion, whereby the multiple equipotential bond connections are arranged at a distance from one another along the longitudinal direction of the wind turbine blade.
9. The wind turbine blade according to claim 7, wherein, the first equipotential bond connection and the second equipotential bond connection comprises or is made from electrically conductive fiber-reinforced plastic.
10. The wind turbine blade according to claim 1, wherein the at least one electrical interface comprises metal or is made from metal.
11. The wind turbine blade according to claim 1, wherein, the at least one electrical interface is electroconductively connected to the electrically conductive fiber-reinforced plastic of the at least one main down conductor by a metallic conductor connection comprising metal or being made from metal.
12. The wind turbine blade according to claim 1, wherein, a longitudinal portion of the wind turbine blade in between the tip portion and the root portion and in between the two spar caps does not comprise a metallic down conductor comprising or being made from metal.
13. The wind turbine blade according to claim 1, wherein the at least one spar web or the wind turbine blade does not comprise a metallic down conductor comprising metal or being made from metal.

14. A wind turbine comprising at least one wind turbine blade according to claim 1, wherein the lightning protection system of the at least one wind turbine blade is electroconductively connected to a lightning protection system of the wind turbine.

15. The wind turbine blade according to claim 1, wherein the two spar caps are the only down conductors of the wind turbine blade.

16. The wind turbine blade according to claim 1, wherein the wind turbine blade does not comprise a metallic down conductor arranged between the two spar caps between the tip and the root of the wind turbine blade.
